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Systematising the Practises and Recommendations of Ukrainian Institutions with Damaged and Destroyed Research Infrastructure: Methods of Calculations of Losses, Recovery Needs, Fundrasing, and Feedback on Data Collection

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White Paper

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Summary

This white paper describes the current situation around the infrastructure destroyed as a result of the full-scale invasion of the Russian Federation in Ukraine in February 2022. The deliberate destruction of this type of infrastructure indicates the enemy's strategy and tactics aimed at harming the prospects of Ukraine for developing its science, innovation, and economy. In this regard, it is crucial to safeguard and preserve research infrastructure (RI) during the war, while documenting and evaluating the damages to record Russian crimes and estimate the costs required for Ukraine's recovery and collection of reparations from the aggressor.

This white paper has three objectives:

1. To systematise information on the methodologies of assessments of damages, based on the data obtained directly from the affected institutions;
2. To provide practical recommendations from the lessons learned by the affected institutions regarding assessment of damages, needs for recovery, and fundraising;
3. To facilitate the exchange of information between the affected institutions and the interagency working group that is charged with preparing reports on damages inflicted on the science sector in Ukraine as a result of the Russian aggression.

The main methods of data collection from the affected institutions were a Google Forms questionnaire and in-depth semi-structured interviews.

This text explores the challenges of calculating the funds required for reconstruction, such as the conditionality of figures and deficient calculation methods. The task of calculating the funds needed for reconstruction is even more difficult due to the lack of methodologies for such calculations and the ambiguity of answers to the question: Should everything that is damaged and destroyed be recovered? In what form should it be recovered? When should restoration begin? What funds should be used and under what conditions? What are the prospects for using the restored RI? Institutions with damaged and destroyed RI find fundraising problematic mostly due to them being governmentally funded entities characterised by a lack of tradition and practice of consistent fundraising efforts from other sources, including funds from sponsoring donors.

This white paper will be useful to the Ukrainian and foreign audiences of researchers and analysts who study the issues of damages to research infrastructure, as well as those involved in policymaking on recovery and development of Ukraine.

Problem

Research infrastructure (RI) refers to the equipment, resources, and related services used by the scientific community to conduct research[1]. Since the beginning of the full-scale invasion on 24 February 2022, scientific institutions and higher education institutions (HEIs) in Ukraine have been targeted by Russia. RI is a type of civilian infrastructure essential for maintaining and developing the country's scientific potential, which the actions of the enemy are putting at risk. According to the February 2023 report on damage to property and facilities of scientific and higher education institutions, published by the Ministry of Education and Science of Ukraine (MESU), around 35% of the RI was damaged and destroyed[2]. 116 institutions were partially damaged and 4 were completely destroyed, while 11 institutions are located in the occupied territory and are not able to provide the data. As of November 2023, real-time mapping of damages conducted by the Junior Academy of Sciences of Ukraine (JASU[3]) revealed that 168 institutions suffered damages (an increase by 52).

The number of damages is growing, and since the war is ongoing with no clear end in sight, it is impossible to predict the amount of additional harm that will be inflicted on research institutions. The deliberate destruction of this type of infrastructure indicates the enemy's strategy and tactics aimed at impeding the progress of science, innovation, and the economy of Ukraine. In this regard, it is crucial to protect and preserve RI during the war, to document and assess the damage caused, as well as to record the Russian crimes and prepare cost estimates necessary for the reconstruction of Ukraine and recovery of reparations from the aggressor. To date, there are several bilateral and multilateral plans to rebuild Ukraine; however, they prioritise civilian infrastructure over scientific facilities.

This white paper presents the current state of the problem of destroyed infrastructure as of November 2023 and has three objectives:

1. To systematise information on the methodology used to calculate losses, based

[1] URL: *https://mon.gov.ua/ua/nauka/nauka/derzhavni-klyuchovi-laboratoriyi*.
[2] URL: *https://data.consilium.europa.eu/doc/document/ST-10034-2023-INIT/en/pdf* with reference to *https://drive.google.com/file/d/1cpn2M9_v7SK9idkiU42h1j9y8s2CFDAt/view*
[3] Transdisciplinary cluster of scientific and educational resources of the Junior Academy of Sciences of Ukraine. URL: *https://polyhedron.ulif.org.ua/*.

on data obtained directly from the affected institutions;
2. To provide practical recommendations from the lessons learned by the affected institutions regarding the assessment of damages, needs for recovery, and fundraising;
3. To facilitate the exchange of feedback between the affected institutions and the interagency working group to prepare reports on damages inflicted on the science sector in Ukraine as a result of the Russian aggression.

This work is a follow-up of the white paper Forming a Database of Damaged and Destroyed Research Infrastructure[4] as part of the previous stage of the Science at Risk project delivered by NGO Kunsht. The first white paper (1) presents a study of the problem of collecting information on damage and destruction of RI and (2) based on an in-depth study of eight cases, systematises the data on damaged and destroyed RI in the form of a table, which serves as a proposal for further development of a database for systematic collection and fusion of information.

The second white paper focuses on methods for calculating losses and recovery needs. Please check the link to the Science at Risk[5] platform to find out more about the database of damaged and destroyed research infrastructure, as well as information on individual cases reviewed.

The quotes from the interviews are provided verbatim[6] for illustrative purposes and in order to provide a space for the affected organisations to directly communicate their experiences and challenges caused by the damaged RI.

[4] URL: *https://scienceatrisk.org/uk/whitepaper/bazy-danykh-poshkodzhenoi-ta-zruinovanoi-doslidnytskoi-infrastruktury*.
[5] URL: *https://scienceatrisk.org/uk*.
[6] Disclaimer from the translator: The word 'verbatim' here refers to the original quotes in Ukrainian. To improve readability, the quotes have been adapted during translation.

Methodology of research

The project was taking place from April to December 2023. The first step involved reviewing the literature on infrastructure ethnography, crisis management, and selected cases of disasters in research organisations. Subsequently, the authors examined the official sources, searched for media records, and contacted researchers directly to gather information necessary to form a sample of research institutions and HEIs whose RI was affected by the war (see Annex 1).

Since May, to improve the effectiveness of work and synergy, the authors of the white paper have been collaborating with the United Nations Educational, Scientific and Cultural Organization (UNESCO) Category II Centre — the Junior Academy of Sciences of Ukraine (JASU). At that time, JASU was preparing a report for UNESCO on the damages caused to the science sector in Ukraine as a result of the Russian aggression. They were also developing the platform "Damage from Russian Aggression: The National Academy of Sciences of Ukraine".

Additionally, since May, one of the authors, acting as a representative of the Science at Risk project, has been working closely with the interagency working group established a month earlier to prepare a report on the damages caused by the Russian aggression to the science sector in Ukraine (the group is chaired by Deputy Minister of Education and Science of Ukraine, Denys Kurbatov). In July, the working group approved two tables: "Information on Equipment/Facilities of Higher Education and Research Institutions Damaged or Destroyed in Connection with the Armed Aggression of the Russian Federation'' and "Information on Real Estate of Higher Education and Research Institutions Damaged or Destroyed in Connection with the Armed Aggression of the Russian Federation". In July and August, these tables were completed by institutions of various subordination levels that own research infrastructure (RI).

The interagency working group requested support from the Science at Risk project implemented by Kunsht to gather feedback on these tables. Consequently, the main work by the white paper authors in close cooperation with the institutions whose research infrastructure was damaged or destroyed did not start until late August and continued till late November 2023.

The sampling was based on the population of institutions that suffered damage or destruction. Specifically, 32 institutions were chosen as an approximate representation of a quarter of the

total number of 116 institutions, as specified in the MESU report[7]. To track the dynamics, the sample was limited to the eight institutions that had participated in the first phase of the project. When forming the sample, the authors aimed to include the most diverse range of institutions possible, considering factors such as geography, jurisdiction, disciplinary profile, degree of damage, and types of damaged research infrastructure. The data were collected via a Google Forms questionnaire and in-depth semi-structured interviews. To supplement and clarify the information provided in the questionnaire, in-depth interviews were conducted with the heads or authorised representatives of the selected institutions The lists of questions used in the questionnaire and in-depth interviews are provided in Annexes 2 and 3.

The responses to the questionnaire and interview questions were coded and analysed to summarise the findings that are presented in the main body of the white paper.

Problems encountered during collection of data

The authors aim to inform reader about the challenges they faced when collecting information from the damaged institutions. This is essential for a better understanding of the findings presented in the main body of the report. Additionally, it may be helpful to those planning to conduct similar research during the times of war and crisis.

The official communication with the sampled institutions commenced with the drafting and sending of appeal letters to their administrations on behalf of the head of NGO Kunsht. Reminders were subsequently sent via emails and phone calls during the first month; however, the questionnaire response rate was quite low. 19 institutions duly completed the questionnaire and provided answers during the in-depth interviews. After giving preliminary consent, seven institutions either refused to cooperate or ceased communication completely.

To provide a different option for five of those institutions, the questions from the questionnaire and the interview were merged, prioritised, and reduced to the total of seven questions. It was observed that some institutions showed less enthusiasm for filling out the questionnaires or participating in the interviews, compared to the first phase of the project which took place between November 2022 and February 2023. Among the reasons of refusal to participate in the study, the institutions cited various bureaucratic constraints, including maintenance of separate records of damage by different structural units, managers being abroad, lack of time, fear of accountability for providing specific figures, possible leakage of sensitive information on damage to the enemy, and the risks of such information going public or semi-public. Furthermore, three institutions failed to provide an explanation and did not respond to any attempts to contact them via their provided points of contact.

To maintain the sample size of 32 institutions, two additional organisations were contacted, which filled in the questionnaire and participated in the interviews.

As this white paper was being completed, five institutions had not yet delivered their questionnaire although no formal refusal to provide data had been received from them. Once they respond, their information will be processed and added to the Science at Risk[8] database of damaged RI.

To minimise the aforementioned issues, it may be helpful to first communicate with the individuals employed by the organisations you intend to contact. Furthermore, we recommend that you formulate your inquiries as concisely as possible and in a manner that gives the respondents the easiest option for answering questions.

[7] URL: *https://drive.google.com/file/d/1cpn2M9_v7SK9idkiU42h1j9y8s2CFDAt/view*.
[8] URL: *https://scienceatrisk.org/*.

Data collection results

Characteristics of the institutions that filled in the questionnaire

As of early December 2023, 24 institutions submitted questionnaire answers via Google Form. Two institutions provided combined responses to the questionnaire and interview questions by phone.

Among these, there is one private organisation, and the other 23 institutions are state-owned and subordinate to the following:

- National Academy of Sciences of Ukraine (NASU) — 10; Ministry of Education and Science of Ukraine (MESU) — 9;
- National Academy of Agrarian Sciences of Ukraine (NAASU) — 2; National Academy of Medical Sciences of Ukraine (NAMSU) — 1; Ministry of Culture and Information Policy of Ukraine (MCIP) — 1;
- State Agency of Ukraine on Exclusion Zone Management — 1; Kherson Regional Military Administration — 1; Zaporizhzhia Regional Council — 1.

Five of these institutions remain in the occupied territory.

The research infrastructure of these institutions may be differentiated by type as follows: facilities, devices, and equipment — 8; civil infrastructure — 4; archives — 2; communication technologies — 1; data collections and repositories — 4; all of the above — 3; musical instruments and library — 1.

Six institutions reported that their RI includes facilities with the national heritage status[9], while three of these institutionsare centres for shared use of scientific equipment[10].

The types of damage include destroyed devices, broken windows and equipment, stolen equipment and data storage devices, and loss of plant and microbial collections due to the inability to maintain appropriate temperature conditions. Additionally, the RI of three institutions is located in the temporarily occupied territory. Most organisations do not provide information on damage or destruction on their websites, in some cases due to security concerns. However, 11 organisations do communicate their losses and needs on their websites.

[9] URL: *https://zakon.rada.gov.ua/laws/show/723-2016-%D0%BF#Text*.
[10] URL: *https://zakon.rada.gov.ua/laws/show/703-2022-%D0%BF#Text*.

13 institutions, which represent almost half of all respondents, reported that their RI could be restored. Seven institutions found it difficult or impossible to answer this question. Four institutions indicated the possibility of partial recovery, while one institution reported no recovery was possible. 13 institutions continue to operate after the damage, seven institutions continue partial operation, while six institutions were forced to cease their work completely.

The estimates of losses and recovery needs range between UAH 60,000 and UAH 1,000,000,000. However, five institutions lack the capacity to record losses, while one institution has its accounting maintained by different structural units.

The following section provides information on how the institutions addressed the challenging task of calculating losses.

Methods for calculating losses caused by damages

Some of the surveyed institutions have not yet conducted damage assessments for their damaged or destroyed RI due to the following reasons:

1) their location in the combat zone;
2) mined territory;
3) lack of funds for commissioning an expert opinion.

Two of the affected institutions reported that they had not assessed the damage. In one case, the material damage was only estimated for the windows and frames of the building, as the institution was unable to conduct a professional assessment. One of the key problems with the assessment was the lack of funds to pay for the expertise. Additionally, the shortage of certified experts in specific areas and the challenges related to bidding procedures further complicated the issue:

"We invited experts [from National Scientific Center 'Hon. Prof. M. S. Bokarius Forensic Science Institute' of the Ministry of Justice of Ukraine]. This institution deals not only with medical cases but also with damage assessments. [...] Its experts have the ultimate authority and are often consulted for court decisions, among other things, so they are very competent in these matters. However, we were not able to procure their services as it would cost approximately UAH 180,000, which was beyond our budget for this year [...] If we approach the Bokarius Institute next year for an assessment and they quote a price of over UAH 200,000, we will have to initiate a competitive bidding process and host a public tender. Such a process typically lasts between one and a half to two months. And there is no guarantee that the Bokarius Institute will win the tender." (2)

The lack of the required experts with narrow specialisation in Ukraine is especially pressing. This also concerns foreign professionals who may have limited or no access to the objects of assessment due to travel restrictions imposed by governments or organisations for the areas where military operations are ongoing.

"While the damage to the infrastructure can be assessed objectively, the damage to the equipment becomes a problem as it requires repair. We need foreign specialists to come here but they are not allowed to come due to security reasons. Alternatively, the equipment might be sent to them for repair, but they cannot say how much it will cost." (6)

Institutions whose entire RI is currently in the occupied territory face acute problems as they cannot apply any methodology to calculate losses without access to their RI. Consequently, they report that they have lost everything, and it is irretrievable. The situation is further complicated by the fact that all documentation on the RI remains under occupation as well. Therefore, it is impossible to calculate the cost of this RI. In one case, a museum-reserve with RI continues to operate in the Ukrainian legal field, despite being in the occupied territory, 10 km from the front line. However, the only personnel present there are guards, and not

researchers, making it impossible to conduct an assessment.

Some institutions in the de-occupied territories still claim to have RI but are unable to assess its current state. Moreover, there is a high likelihood that this RI will need to be destroyed for security reasons.

"Some archives, such as the one in Vysokopillia (this information is no classified), have been badly damaged by shelling. Despite attempts to demine it, there are so many tripwires that, according to the State Emergency Service of Ukraine, it would be easier to blow it up than to demine it. We are not able to even retrieve the Vysokopillya archive because we do not have any physical access to it." (14)

The relocated institutions whose entire material resources remained under occupation have found themselves in an exceptional situation. To our knowledge, the RI of some of these institutions has sustained no significant damage or destruction. Nevertheless, it remains completely unavailable for use, and its future is justifiably uncertain:

"Based on the experience of Kherson HEIs, it is impossible to predict how much will be left there after recovery [de-occupation]." (15)

Institutions are unable to conduct damage assessments while in occupation, even when international organisations have the budget to arrange monitoring but cannot do so as such monitoring requires an on-site visit (15).

It is worth mentioning that unregistered museums at educational institutions have been excluded from the list of destroyed and damaged RI. Additionally, the institutions whose RI is sprawled across a large territory, such as the exclusion zone in Chornobyl or the territory of a radio observatory, require transport vehicles for their research activities. As their transport was either destroyed or stolen during the occupation, it cannot even be directly registered as part of their RI. The same problem is reported regarding hostels near research institutions in remote areas, where researchers can stay overnight to ensure the functioning of the RI and conduct research.

The institutions that calculated their losses stated they followed the methods outlined in the following regulations: Order on Approval of the Methodology for Determining the Damage and Amount of Losses Caused to Enterprises, Institutions, and Organisations of All Forms of Ownership as a Result of Destruction and Damage to Their Property in Connection with the Armed Aggression of the Russian Federation, as well as Loss of Profit from the Impossibility or Obstacles to Conducting Business Activities[11] of 2 December 2022, and Resolution on Ensuring the Functioning of the State Register of Property Damaged and Destroyed as a Result of Hostilities, Terrorist Acts, or Sabotage Caused by the Armed Aggression of the Russian Federation against Ukraine of 13 June 2023[12]. One institution followed the methods outlined in the Resolution on Approval of the Procedure for Determining the Damage and Losses Suffered by Ukraine as a Result of the Armed Aggression of the Russian Federation of 20 March 2022[13].

Several institutions assessed the total losses based on market prices as of the date of the inventory, while others assessed it based on their original cost, mainly using internal financial data. One institution applied the methodology for determining the book value of collectible specimens dated 20 November 2003, along with the corresponding indexation of value. The relocated university also used the accounting certificate drafted on 10 June 2022 for the losses caused by the Russian military invasion, and the value of property (real estate,

[11] URL: *https://zakon.rada.gov.ua/laws/show/z1522-22#Text*.
[12] URL: *https://zakon.rada.gov.ua/laws/show/624-2023-%D0%BF#Text*.
[13] URL: *https://zakon.rada.gov.ua/laws/show/326-2022-%D0%BF#Text*.

movable property, and other items or property) located on the university premises.

The institutions located in the occupied and de-occupied territories faced challenges in drawing up damage reports, as their report templates did not account for lootingof RI objects by the aggressor:

"*The most interesting thing is that there are no specific legal documents and we ourselves developed a deed , as we had come to understand that none of our methodological documents contains any deeds on looting of the property. There is a template for literally anything else — loss, damage, accession, separation, but not looting. That is, our legislator did not anticipate those possibilities.*" *(14)*

The problem of evaluating RI has become even more challenging for the archival institutions as most of their documents, which are definitely part of their RI, have never been assessed before and are only recorded in the books by storage units without specifying their book value:

"*What methodology is (or can be) used to evaluate the original archival documents that were lost? [...] When acquiring insurance for unique documents, a special methodology is employed for their evaluation. Unfortunately, there is no concept of monetary equivalent for the documents stored in the National Archival Fond. These documents are considered historical and cultural heritage and are not material, so their value cannot be estimated [...] And since we only acquire insurance for individual storage units, for instance, when we take them out for a museum display... There is also a certain group of unique documents that is subject to mandatory annual insurance. As far as I know, no archive has recorded this type of cost.*

When we look at a balance sheet of the archival institution as it is, we see a building, some movable property, library property (it can be valued because it is on the balance sheet, there is a clear cost per copy). But manuscripts are not accounted for. And I am not aware of a mechanism to do this. I cannot say that it does not exist, but I do not know it. I only know that there is an assessment mechanism for insurance purposes, but it cannot be used to cover all the archival funds... We had under a million storage units, 970,000... That is, the archive would collapse if we kept all the accounting records of them. We never had the need to do this work. Moreover, any accounting figures require indexation to take into account their depreciation and wear and tear..." *(14)*

Assessing the value of a unique RI presents significant challenges, particularly when dealing with one-of-a-kind library and archive collections or unique equipment impossible to be assessed. For instance, specialised instruments were created over 30 years ago at one of the Chornobyl laboratories to meet the specific demands of the site. The original developers have since passed away, which makes replication impossible.

The vast majority of institutions with damaged and destroyed RI have obviously failed to calculate their intangible or indirect losses or lost profits. Some referred to a lack of methods for such calculations, while others named lack of time or motivation as their reason for not doing so. A representative of a private RI emphasised that assessing the loss of reputation was impossible. They further stressed it was not the reputation of a single institution and its professionals that had been affected. Ukraine as a research location had lost the trust of the international community due to security reasons and ethical concerns regarding the ability to conduct research during the war.

Several institutions emphasised the conditional nature of their reported damages, as many of the damage consequences only become apparent over time:

"*There may be delayed consequences: plants that were damaged at the time may die later.*" *(13)* "*After the strike in 2022, it seemed that only 60 windows were damaged. But [...] in 2023 the*

consequences of the strike have become more severe. The building, which is 200 years old, is in the centre of the city. It is not a historical monument." (1)

For certain institutions in the de-occupied territories, the amount of damage over time may decrease, or even be reversed, if they recover some of the property that was stolen by the occupiers:

"*The work has been going on for almost a year now, and we are constantly updating information. Some property is found, the police find something stolen from the archive and bring the items back from other institutions in Kherson. It is an ongoing process that has not yet been completed.*" *(14).*

Institutions that participated in the previous phase of the project generally failed to use this time to develop new methods for the calculation of losses:

"[...] there was no need to make any decisions. We work reactively, responding to various needs as they arise. When the windows were broken, we calculated the cost, purchased replacements, and installed them. How much was lost? Maybe we will calculate it at a later time." (11)

Perhaps the only institution that has developed its own assessment methodology and continuously improves it is the M. M. Hryshko National Botanical Garden of the National Academy of Sciences of Ukraine:

"[...] Like before, we now have to follow the objective methods of assessing green plantations (excluding the forestry industry, where such plantations are considered a commodity, timber). Our green plantations are valuable plants. It is certain that there are no established methods in Ukraine for determining the book value of specific collectible plants. Each institution develops its own methods for its collections independently.

In 2003, we developed separate methodologies to assess the book value for open ground plants and greenhouse plants. The challenge is, there are a lot of approaches that can be used here: maintenance costs; coefficients of plant uniqueness, decorativeness, and so on. Therefore, for a more or less realistic assessment, we chose the simplest possible approach where each curator could calculate the book value of the plants using a formula. As for the greenhouse collection, we accounted for the age and uniqueness of plants. We divided the plants into two groups, trees and herbs. We came up with a formula where the sum is equal to the uniqueness factor multiplied by the age of the plant.

After 2003, we made several attempts to improve this methodology, and the improvement efforts are still ongoing. The most recent revaluation occurred in 2013, when the book value of the collection exceeded UAH 3 million, and then further revaluation took place based on the fluctuations of the dollar exchange rate... The 2003 methodology was formalised in a decision of the commission approved by the director. Unfortunately, it has not been published. Practically, it is an internal document of our institution, although the Council of Botanical Gardens and Arboretums of Ukraine has made inquiries in an attempt to make that document available for public use. But so far it remains this way." (13)

The most systematic approach to calculating the widest possible range of damages was taken at the relocated Mariupol State University, whose RI was partially destroyed and partially left under occupation:

"The first thing we did was to hand over all the documents to law enforcement agencies and file a claim against Russia for damages caused to the State of Ukraine, specifically for the destruction of property and infrastructure of Mariupol State University, including the loss of profit by the university. The total amount claimed is over UAH 300 million, which is the book value of our property and technical facilities. When Russia loses, we will go to court to demand compensation in the form of reparations [...] We calculated the book value of all property, land,

and lost profits. Law enforcement agencies there have this methodology. A criminal case has been opened, we have declared this amount and informed the Ministry of Education and Science" (9)

To summarise the information presented in this section, estimating damage and loss resulting from destruction and damage is a difficult task due to the novel and unique nature of the problem, lack of approved methodologies, the specifics of different types of RI, and the location of affected institutions at the time of the assessment. In some cases it may be impossible to determine the value of collateral or intangible losses in addition to direct losses. When talking about collateral damage, the respondents referred to the extinction of collections of microorganisms or botanicals due to power outages. The affected institutions also listed intangible losses, such as lost human lives, time, reputation, and opportunities.

Forming a recovery budget

Almost all interviews with representatives of institutions whose RI was damaged or destroyed emphasise the fundamental differences between the cost of destruction/ damage and recovery needs. The reader should be aware that the Ukrainian RI is characterised as outdated and with underfuned efforts for its modernization. For example, one of the institutions noted that certain RI equipment is listed in the accounting records as having a value of "0", or is written off or depreciated, while in reality it is still being used for research.

It was observed, with a certain regularity, that most institutions prioritise restoration and modernisation in order to "build back better than before", rather than calculating the cost of purchasing similar or repairing damaged equipment:

"One needs to separate the cost of damage from the cost of recovery. The latter should be much higher. Moreover, we do not know this cost yet, it does not have the upper limit. We will understand in a year or two." (1)

A number of institutions have assessed the market value of equipment and repairs for their RI facilities and provided estimates. However, these estimates are only valid as of today and may change in the future:

"We have a defect report. This means that a technical inspection has been carried out. How much will the work cost? I do not know... We have estimated [RI] as well. A new grand piano costs UAH 6 million. We have 69 of them, and they were quite old even before the war. Now they will be exposed to the cold for the second winter straight [due to damages no heating has been available in the building for two consecutive years, and the windows have been left broken for a long time]. So we are talking about 69 pianos multiplied by six million." (10)

In accordance with the law[14], the cost must be established on the basis of the project documentation and related expert opinions. This requirement may conflict with the "build back better than before" approach in some cases.

In many cases, the recovery itself requires special equipment, without which losses can be much more extensive:

"For example, the swimming pool that was hit by a hostile projectile... The state once spent UAH 15 million on it. So what makes the situation worse? We cannot drain the pool because we do not know the groundwater levels after the Kakhovka dam was blown up. One of its walls has been punctured, and there is a lot of debris there. So we cannot drain the water because the pool basin may crack, and this is the most valuable part of

[14] URL: *https://zakon.rada.gov.ua/laws/show/326-2022-%D0%BF#Text*
https://zakon.rada.gov.ua/laws/show/624-2023-%D0%BF#Text
https://zakon.rada.gov.ua/laws/show/z1522-22#Text

the pool. This means we need special pumps to drain and sieve the water to remove the debris. There are robotic [pumps] that could do this, so certain work could already be done to help." (3)

At that, not all the RI can be recovered:

"How would you compensate? How can you compensate for the loss of a 15th century book? There is no way, as you cannot recover it. You can only record the fact that it is gone. Merely form a database." (1)

"There is practically no way to restore unique collections, especially if they are in the Red Book and were imported during the Soviet times.Today, no one would allow some of these Red Book plants across the borders." (13).

Bureaucratic overregulation was also cited as one of the obstacles preventing recovery:

"There are financial issues. For example, we have money and can rebuild everything without asking for anyone's help. But there is the Treasury, tenders, and restrictions on procurements without tender procedures. We need a can of paint, and we have already spent the funds allocated for this specific [Classification of types of economic activity], so we need to hold a tender. There are so many delays and obstacles in this processthat we need some kind of a government decree that allows... If you need a box of nails, do you have to hold a tender? Two proposals were submitted, then it stopped, the tender was cancelled, no one submitted a proposal... It is horrible. If it is about recovery, there should be a bank account, a payment, and then we receive these materials in one hour instead of waiting for months. It is a nightmare. We are constantly walking the line: We need something, but we have no right to get it, so we need to be creative about it. What does it mean? Someone will come tomorrow and say, 'Aha, you guys are putting something in your pockets here.' Therefore, we need permission for direct financing of these needs at our own expense. At least for the period of martial law. As for the equipment, what are we even talking about?" (4)

Therefore, calculating the cost of recovery is just as challenging as calculating losses. The RI recovery estimate drafted with the "build back better than before" approach in mind is not a static document. Estimates may become outdated due to price changes, inflation, additional damage, or the damage that occurs naturally over time. Moreover, certain collections, such as microorganisms that have been cultivated for decades, or valuable archival documents, are impossible to restore.

Fundraising:

The reader should be aware that the model of competitive funding of research and raising funds from external sources is a relatively new practice in Ukraine. Historically, just like it occurred during the Soviet times, most research institutions were funded directly from the state budget allocated for science, which has been below the target of 1.7% of GDP since Ukraine gained independence. In 2019, it fell to a record low of 0.29% of GDP[15].

15 out of 25 institutions reported that they have neither formal nor informal algorithms for fundraising. This means they lack formal or informal algorithms for raising, using, and reporting on the funds that would enable them to achieve a specific goal for the development of their RI. These institutions mostly implement ad hoc fundraising activities that cover immediate needs. Only seven institutions indicated that they have formulated fundraising strategies.

Additionally, state-owned institutions often encounter issues when it comes to charitable aid:

"There is this Law of Ukraine 'On Prevention of Corruption' says that a state-owned institution is

[15] URL: *https://data.worldbank.org/indicator/GB.XPD.RSDV.GD.ZS?locations=UA&view=chart*.

funded from the state budget and cannot receive any type of financial aid as this is considered an act of corruption. Therefore, being a state institution, we are not eligible for any such aid. It should be specific or local grants... However, our eligibility for most grants is limited as they are only available to nongovernmental organisations. We are only allowed to collaborate with the UCF [the Ukrainian Cultural Foundation]. We have received some aid primarily from fellow archivists. This property is accounted for outside the fund, and we are not allowed to receive payments or hold a foreign currency account." (14)

However, some of the organisations interviewed were able to raise funds through established charitable foundations by appealing to partners, colleagues abroad, former employees, and alumni. According to a respondent from a university, the institution did not have a strategy. However, after local media coverage of the damage sustained by the university, banks and entrepreneurs from across the city reached out and covered immediate reconstruction needs, such as repairing the roof and replacing windows.

The Radio Astronomy Institute of the National Academy of Sciences of Ukraine houses the world's largest low-frequency radio telescope, which is considered a national heritage object. Despite the war and the damages, the institute is working on a plan to modernise its sections — unique low-frequency instruments that secure Ukraine's global prioritised position in low-frequency radio astronomy. This work is covered by both theNASU funds and grants provided by international partners.

Another example of an innovative approach to fundraising was provided by a relocated university:

"We made a technical specification of what should be on the new campus of Mariupol State University in Kyiv. We called for proposals from students of technical specialities. We will select the winners and implement the winning project. Following this, design estimates will be procured. We understand that we need a significant budget for the construction of dormitories and hostels. We are actively seeking funding for this, communicating with Ukrainians abroad and other partners." (9)

Additionally, the university is adopting a comprehensive and multi-level approach to future funding. It has filed a lawsuit against the Russian Federation to claim damages. However, the institution does not solely rely on the lawsuit for recovery but is also actively engaged in fundraising.

"Our university is launching a new digital platform integrated with Diia [a web portal for accessing public services]. We have posted special fundraising videos there designed for various countries, including the Ukrainian-speaking audience. We have registered a dedicated fund, and our foreign partners raise funds on our behalf. We reach out to embassies, non-governmental organisations, legal entities, individuals, and Ukrainian businesses... The new platform will offer electronic payment systems to help the university." (9)

Despite being proactive in raising funds from foreign donors, some organisations reported difficulties in obtaining funds and equipment during the war. They stated that their international partners understand the situation is long-term and any support is viewed as something to be provided in the future:

"We are all sensible, and I understand that it is not the right time. I don't think anyone is going to buy grand pianos for Kharkiv and put them here right now. We have basic preliminary agreements with donors from the Netherlands, they say, 'We will be happy to help you, we will buy a certain number of instruments and provide them after the victory. We will not send anything to Kharkiv now.'" (10).

Another problem that was specifically highlighted is the customs clearance of

humanitarian aid in the form of equipment delivered for research institutions. In some cases, volunteers transported small equipment as private individuals, without declaring its value to the customs as there is no specific legislation to regulate this issue. In this way, they delivered functioning equipment of a specific type which is strategically important for the industry.

Therefore, Ukraine still lacks broad competitive funding for science, as well as excellence-based funding. Prior to the full-scale invasion, only a small number of institutions had experience in raising extra-budgetary funds, developing an institutional fundraising strategy, or collaborating with businesses or other organisations, including foreign ones, through partnerships. The fundraising methods employed by the affected institutions during the war were primarily focused on addressing immediate needs and were sporadic rather than strategic, consistent actions to achieve specific goals.

Conclusions

The in-depth study of 32 cases of institutions with damaged or destroyed RI has led to the following conclusions.

The estimates of losses suffered by institutions with damaged RI are uncertain due to the imperfect methods of their calculation. These estimates are accompanied by many difficulties due to the following:

- extreme circumstances of the war;
- lack of precedents and previous experience in organising such estimates;
- lack of flexibility in the system which imposes many bureaucratic restrictions;
- insufficiency or lack of methods for calculating damages of specific types of RI;
- lack of specialised experts and funds for expert assessments. Furthermore, the consequences of damage often only become apparent much later, requiring a recalculation of the amount of damage.

Calculating the funds required for recovery is an even more challenging task. Firstly, unlike the calculation of damages, there is no established methodology for determining the costs of recovery. Secondly, there are no definitive answers to the questions of whether everything damaged and destroyed should be restored. In what form should it be restored? When should restoration begin? What are the conditions and funding options? What are the prospects for using the restored RI?

The challenges associated with fundraising by institutions with damaged or destroyed RI are primarily due to the specifics of financing for budgetary institutions, as well as the absence of established practices and traditions for systematic fundraising from alternative sources, such as donors. Institutions often lack strategic approaches to fundraising. Along with this, international donors and partners are ready to help with recovery. However, this aid may be subject to certain reservations, conditions, and delays until the end of hostilities. The challenge lies in the alternatives offered by China as a potential donor for reconstruction. Unlike European, American, and Japanese donors, the Chinese are willing to provide funds for reconstruction without waiting for the end of the war, imposing a list of requirements, or burdening this aid with bureaucratic obligations. Nevertheless, this primarily applies to RI that is strategically important for the country's economy and security.

The accounting of damaged and destroyed RI is complicated by the following factors:

- institutions may lack motivation and understanding of how filling in the necessary tables with concrete figures might help this particular institution;
- lack of communication and feedback from requesting authorities who provide requirements for filling in the tables;
- institutions may be heavily burdened with requests for damage and destruction reports;
- the affected institutions may have security concerns about disclosing these data publicly.

Regarding generalisations about openness, some institutions were willing and grateful to be contacted and talk about their damaged and destroyed RI. First, they hoped to find donors for reconstruction; second, to have their problems and challenges heard; and third, to draw more attention to the problems of the affected research institutions in Ukraine.

Others opted not to share information, either due to concerns that it may be used by the adversary, or to prevent alienating their international partners who may doubt the ability of the affected Ukrainian institutions to meet the commitments.

It is important to consider that the problems with the calculation of damage, restoration needs for RI, and fundraising are closely linked to the human resource issues, including loss and lack of staff, increased workload, psychological problems, and more. For further information on this topic, please refer to the Science at Risk white paper on human resources.

Recommendations

Recommendations for calculating losses, recovery needs, and fundraising, based on lessons learned by the affected institutions

Most affected institutions did not formulate any systematic recommendations to improve methods of calculation of losses and recovery needs. The Institute of Mathematics of the National Academy of Sciences was the only institution to propose a radical change in the approach to such calculations, formulating a logical and reasoned algorithm:

"[...] The only way to assess how much it will cost is to make an estimate. This costs money, and the price is constantly changing. The estimate is valid for a month or two, and then you have to make a new one. Prices have changed along with the market situation. This means the institute has to spend more money every few months to update these figures. This is impossible. Otherwise, they are just a formality.

Therefore, the most logical option in this case is to roughly record all the approximate damages, create a preferably sensitive, private database, and use it to contact sponsors so that they could make their own cost estimation. Professional expertise should be paid for by sponsors concerned, and these will be the real figures. The sponsor will trust this estimate because they were the one who commissioned it, and then decide whether to fund the recovery or not.

Another benefit of this approach is that the process is dynamic, and the amount of destruction keeps changing. That's why you need an updated estimate only when you find a sponsor willing to spend money. Otherwise, institutions may get stuck in bureaucracy and struggle to close existing cases... The only legal way is an estimate prepared by an organisation that has a licence to do so. Now, if I find a sponsor, I will have to submit a package of documents, including an estimate... Why should institutions develop new methods for calculating damages inflicted on their infrastructure, if someone claims later that the current figures are unreliable? Let us consult experts and they will ensure accurate calculations [...].

Recommendations for recovery: No one will just give money for existing institutions, unless they find a personal sponsor. To get the money, one needs to create new projects. We

could call them centres of excellence. New "crystallisation points". Say, "Okay, the old is destroyed, let's build a new one from scratch." And then submit it as a project and let the international expertise assess how realistic it is [...].

There are situations when an institution is destroyed. Then a new institution should be created in its place, something small, dynamic, and one that is actively working [...]

Regarding sponsors for the reconstruction, someone global, like UNESCO, should start raising money. Let them assess and decide what to restore. Then there will be complete trust. The selected sponsor can then commission an expert assessment and proceed with the restoration. However, this should not apply to absolutely everything because the sponsor may say, "Is there a guarantee we will rebuild the institution and it will operate the way it did before, at the very least?" One must submit a comprehensive project that includes estimates for staff and other relevant aspects... The description should cover the prospects of the project and whether it will have students. [...]

I was told that Poland had applied a different strategy: the State provided 25% of the funding for such successful projects, and the EU provided the rest. However, this was only done if the project clearly described all the prospects. Whoever pays for this should be able to see what will happen next, who will work, etc. [...]

Therefore, it should be a database of damages with photos, etc. + search for donors —> Donors conduct their own assessment (or pay for such an assessment) —> Allocate money to whoever they see fit(1).

To a certain extent, the rector of Kherson State University's vision aligns with the approach proposed above:

"In my opinion, no cash should be handed out. Instead, write out a clear request for what needs to be purchased and delivered, which should then be thoroughly checked. You need to be able to inspect everything. Why? Because it is transparent; otherwise, this means the treasury, dirty publications [in the media], etc. I believe, good name and reputation are more important than anything else in this sense. [...]"
"It is clear we should spend the lion's share of our investments not only on material things. The following principle should apply: For example, you have lost TVs and computers in some classrooms. You applied to the foundation, and the foundation replaced them. No money should be involved. Applications are processed, and then items are procured and delivered. You come, you record it (on photo or video), and you send it. Come, record, send. Make sure it has a geolocation mark to avoid the impression of a fake report. The actual money should be invested in the development of the intellectual, intangible component. This is a key resource that must be developed at universities. [...]"

"How to evaluate? I believe that evaluation should only be conducted by licensed professionals. There are not enough of them for everything. This is what we do in our cooperation with USAID: We submit an application. They call back and ask us to show our situation and needs. We do so. After an interview they say, "All right, you will get the necessary equipment", and we get it. Then they send a specialist to check, but no monetary evaluation actually takes place. They come, see what we have, and confirm we have it. Then they ask whether it will work. This is how it should be, a three-level process. 1) We filmed it and sent it. They checked it out, and if they have any questions, they ask us. 2) Then they deliver what we need. The delivery should be like this: A commission must be present at the door, signing paperwork to confirm receipt and avoid falsification. (I am telling you how we do it.) Next, the accounting department makes an entry in the books based on the certificate of acceptance. That's it. It is now on the balance sheet, and you cannot steal it. After that, the donor may send someone for inspection. After that, requests can be made to openly show the work, on Linkedin, Facebook, or anywhere else,

show everyone how this space is used. That's all, and then nobody questions us. They say, We've seen the activities, we see the kids, we see the students, we are ready to continue funding" [...].

How should it be arranged? *"A list of [damaged] objects. The future sponsor gives us the relevant questionnaires which we will then send back to them, knowing that we have a chance to get the money after negotiations. Questionnaires may have several different blocks, for instance, this is what we need today, for example for a swimming pool. Another block is for the things to be done after the war is over. And the third block is for the things that should be done strategically, for example, in two or three years after the war." (3)*

In this way, the leaders of both the Institute of Mathematics and Kherson State University emphasise the following: a) the priority of calculating costs for recovery needs rather than calculating actual damages; and b) calculations to be conducted directly by the donors willing to invest in recovery, rather than by the institution itself or any hired experts.

At this stage, the State should consider other needs related to the affected institutions. These include building trust, establishing communication, understanding the needs (besides material ones) of the affected institutions, providing support to cover immediate needs and "make their lives easier" (regulating customs clearance of humanitarian aid sent for scientific purposes, and giving priority to demining of their territories). Additionally, the State should consider the research infrastructure as part of the country's innovation system, understand the needs of other actors such as private RIs, RI users, and international partners, as well as integrate RI recovery into the country's recovery plans.

Regarding other practical recommendations from institutions, the following propositions should be pointed out:

- "In the face of rapidly growing demand for expert assessment of damages and destruction, legislative measures should be taken to minimise the risks of "a bunch of commercial entities flooding the expert market to provide biased assessments for a fee. They will write whatever you ask in the assessment." (2) Institutions were concerned that these commercial entities may make more attractive offers during a bidding process, potentially sidelining true experts in a specific area.
- A clear, consistent, and understandable algorithm for calculating losses is required, as people are confused, psychologically overwhelmed by the war, stressed by the unclear situation and the loss of an important support in their lives, i.e. their physical place of work.
- The M. M. Hryshko National Botanical Garden of the National Academy of Sciences of Ukraine is willing to refine the methodology it has developed for assessing the book value of plants: "Would we recommend it for the whole Ukraine? The methodology needs to be revised, adapted for the whole country, and discussed to draw up an agreeable option... We could effectively work with this methodology and then submit it for approval" (13)
- "What needs to be considered during calculation: For indoor plants, the key costs are maintenance and heating. For open ground plants, principal attention is paid to their uniqueness and age. The formula should not be complicated, otherwise people would be overwhelmed by calculations. Another important aspect to consider is subordination. Some botanical gardens are managed by the National Academy of Sciences, while others are managed by the Ministry of Education and Science. Another important thing is to avoid overcharging. Plants may die, and then the costs would be huge. So the answer is that the formula has to be quite simple and generally acceptable throughout Ukraine, regardless of the

departmental subordination of a particular botanical garden." (13)

- "We are a state-funded organisation, so we are not allowed to receive funds from donors" (6)
- "Financial autonomy of universities should be expanded. We face difficulties with paying bills through the state treasury, and we are unable to open special fund accounts in non-state banks. We have worked with all state-owned banks; to be honest, state-owned banks are not the most suitable partners. The foreign partner transfers an amount which is held somewhere for three months before being returned to the donor, incurring a commission and an additional UAH 1,500 charge for responding to our request. This is why we have registered a foundation to be able to hold accounts in non-state banks. Another problem is that the university is unable to cover all expenses. However, we are trying to resolve all the issues." (9)
- "It is nearly impossible to write off the equipment that has been damaged or destroyed. We are a state-funded organisation, and everything is written off through the Cabinet of Ministers. The only items that were automatically written off and transferred were the ones we donated to the war effort at the beginning of the invasion. Otherwise, any device purchased back in the Soviet Union times is very difficult to write off." (6) Therefore, the recommendation is to simplify the write-off procedure (6).
- Difficulties in obtaining aid in kind (such as equipment) rather than cash. "There is a mechanism for state registration of this aid. We have worked out the mechanism for registering more than 200 various equipment units donated by the University of Hull, UK. Today we already have this equipment. So we had to cooperate with the Ministry of Finance, the customs, the border guard service, and register projects in time. Everything is achievable. Naturally, we are ready to share this experience" (9) — "zero tax return" — humanitarian aid for state institutions (29)
- "Humanitarian cargo needs to be registered correctly" (9) + customs clearance (17) (22)
- "Regarding the development of a fundraising strategy, UCU [Ukraininan Catholic University] and the Kyiv-Mohyla Academy succeeded in this. But, in fact, like most universities, we do not have results as perfect as theirs. We are talking more about declarations than real results... We need to move from declarations to real action. We've made a mistake by posting accounts for donations on our website, and in the first weeks of the war we hoped that our foreign partners would send money there. We were very naïve, because even when the pressure was at its highest, we received $100-$200 from foreign partners, and I think that the total aid we received did not even hit the $1,000 mark. So, frankly, in the long run, the university needs to create both regulatory frameworks and a new real fundraising model... The lack of such traditions causes a lot of issues. The problem is that universities in Ukraine are not trusted due to bad communication. Even MESU admits that there are few successful fundraising practices. The idea of an endowment is exotic for Ukraine, given the specifics of the stock market and banking sector. We need to develop trust, when the university not only begs but also gives. This means research and the third mission of universities" (5)

Recommendations based on feedback exchanged between affected institutions and the interagency working group to prepare a report on damages inflicted on the Ukrainian science sector by the Russian aggression

Out of the 26 surveyed institutions, 11 reported they had received and completed the tables titled "Information on damaged or destroyed equipment/institutional facilities due to the armed aggression of the Russian Federation" and "Information on damaged

or destroyed real estate of the higher education institution/research institution due to the armed aggression of the Russian Federation". These tables were compiled by the interagency working group in July 2023. Another 11 institutions neither received nor completed the tables. During the interview, three of them requested a reminder to be sent to the interagency group about their situation and the damage to their RI. Two organisations received the tables but did not fill them in for various reasons, and four did not respond to the request.

One common theme was the observation regarding the constant growth of the amount of paperwork that needs to be done to respond to numerous requests from various authorities.

Most respondents reported that the tables were clear and contained instructions on how to complete them. Institutions provided the following recommendations to the interagency group as feedback:

- Coordinate the process of collecting information and take measures to avoid duplication of requests sent to the affected institutions: "So far we have completed a lot of these tables! From the first day of this phase of the war, we have been filling in various data on damaged infrastructure. They are sent to us by MESU (by its various departments), the regional administration, other services, and NGOs. So we just enter the same information into new tables over and over again." (11). "It would be nice if it was accumulated in one place because we have submitted 200 of these forms to different places, they are only slightly different, and they change. Damage reports. City council, regional council, ministry, another one, and another one, and another. They have only minor differences. Maybe I'm simplifying the situation as I'm not an expert, but I see it this way. For instance, there is the Ministry of Infrastructure that is supposed to be involved in infrastructure restoration, so it should collect information and say, 'We need this, and now you have to do it.' Otherwise, it turns out that you submit the data to many places, and none of them answers: What happens now? I wish there was such feedback. I would like a clear algorithm." (10)
- The institutions sending such tables and questionnaires need to be more specific about the purpose of collecting specific information. Bilateral communication needs to be established. Currently, institutions only receive requests and there is no follow-up communication on how these completed tables are used.

Institutions receive too many documents at the same time and lose the understanding of priorities. *"Sometimes people are slightly frivolous about it as they wonder, 'What is the point of filling them in, what do we get out of this? There will be no serious support from the state anyway.' It is also a matter of trust. It is built when we have feedback" (5). "It is important to help HEIs really understand why they are doing this, so they need real feedback. Not only for submission, but also for MESU to improve their submission of statistical information to higher authorities, such as the Cabinet of Ministers or the Office of the President, then there will be more motivation to submit verified information. In the meantime, there is a risk that some laboratory assistants or secretaries will fill in the forms, and, respectively, the quality of the information will be low" (5).*

- RI and the needs for its recovery should be taken into account not only for the state-owned or municipal but also private institutions.

The following reservations have been provided regarding the format of the tables:

- Universities highlighted the difficulty of making a clear distinction between research and teaching infrastructure. Using laboratories and classrooms for both teaching and research purposes was provided as an example. When collecting

information on the affected universities, one should consider that the university is an organism and an ecosystem, and that one activity should not be separated from the other. Therefore this infrastructure should not be assessed separately. (7) "It is very difficult to give an estimate... We have dual-purpose classrooms and infrastructure facilities, which means they are used both in the learning process and for research. For instance, the swimming pool... The dean of our physical education department had conducted research there for two years. Part of this research was included in his doctoral thesis... Listen, if academic work and research work are not connected, it even looks ridiculous. In the 21st century, a teacher creates new meanings. This is not a school where research has no sense. This is not the case in universities." (3)

- Institutions whose RI remained under the Russian occupation noted that the authorities which send them such requests fail to understand the context of occupation which "looks insulting and cynical [...] And then you shut it downand don't want to complete such a questionnaire." (7) Additionally, it was noted that these specific forms did not contain a column to indicate that the RI was located in the occupied territory.
- Archives and museum collections do not fall under any category in the tables. There is a question about equipment and facilities but nothing about collections. This is further aggravated by the problem that there is no methodology for evaluating collections (14).
- A similar problem arises for institutions with physical data repositories, such as collections (of plants or microorganisms). In the current wording of the tables, it is unclear where to enter information on the damaged collections. Therefore, a recommendation was made by several institutions to add the word "collections'' to the table (13).
- The Vivarium (Institute of Endocrine Pathology in Kharkiv) requires an EU-approved licence and compliance; moreover, the collection of biospecimens that had been nurtured for 15 years was then destroyed by a power outage, making it impossible to calculate losses and estimate costs. Scientists also reported another problem, that even if they use part of the collection that was preserved, it makes the data presentation incomplete and affects the quality of the study.

Recommendations on the frequency of collection of information on damaged and destroyed infrastructure:

"Such information should be collected after the occurrence of damage, as soon as it happens. Ideally, it should be done by a commission with a representative from MESU or some other department. This was how it was done at the beginning of the war. The rectors had a chat where they immediately reported damages. This is how MESU was informed. Collecting information on whether anything has happened at all is not effective." (11) The information should be updated as soon as it changes (9).

The first white paper on the damage and destruction of RI[16] provided recommendations for the establishment of a database, its management and use, and the Communication on the Database. The authors recommend the interagency group to consider the suggestions made in the said white paper and incorporate the recommendations from institutions provided in this white paper in its work of collecting data from the affected institutions. Furthermore, we recommend taking a broader view of RI in the context of the country's innovation ecosystem, establishing two-way communication, and identifying the losses and needs of RI of different ownership forms, as well as identifying the needs of actors such as RI users, international partners of institutions,

[16] URL: *https://scienceatrisk.org/storage/lp/45/582acd08c09e96c1c518b3d34f425f094e7365a7.pdf*.

technicians and engineers who provide maintenance of equipment, and more.

Recommendations to continue the Science at Risk mission

The Science at Risk project will provide practical support to institutions with damaged or destroyed research infrastructure, presented in the sampling, by compiling and publishing data on urgent needs (including non-financial needs) of the said institutions on the website *https://scienceatrisk.org/*, both in Ukrainian and English This will enable potential donors to provide support directly to these institutions.

The affected institutions highlighted requests for the following intangible needs: networking with international partners, capacity-building of scientists (free online courses, including English language courses), information on technology transfer and commercialisation of research, skills in writing grant applications. Private RIs in the medical field should be given an opportunity to publish their achievements in leading medical journals at a reduced fee or free of charge.

The authors recommend that the Science at Risk initiative should continue its mission of connecting potential donors with organisations that have suffered damages and require support.

It is crucial for the affected institutions to know that their time, attention, and effort invested in completing the questionnaires and interviews are reciprocated. Since the beginning of work on the first white paper a year ago, the affected institutions have demonstrated decreasing interest, enthusiasm, and trust. There is a risk that people will lose faith and trust, as some have expressed doubts about contributing to this iteration of the project, pointing out that "some grant projects are done solely to obtain another grant rather than focusing on the stated mission".

Another recommendation is that Science at Risk should take the lead in guiding the shaping science policy practices in Ukraine and raising awareness among decision-makers about the country's innovation ecosystem, its actors, and their needs. It is equally important to facilitate the "science for policy" exchange during wartime and the future recovery. The accomplishments of the NAASU institutes and medical research institutions, particularly in preventing public health issues caused by long-term war stress and PTSD, are noteworthy. These areas have strategic importance for ensuring food policy and public security.

The reports covering individual institutions, currently published on the NGO Kunsht website in Ukrainian, should be translated into English to raise awareness among international audiences.

Annexes

Annex 1. Sampling of institutions

##	Name of institution with damaged or destroyed research infrastructure	Filled-in questionnaire	In-depth interview	News report about the institution	Combined questionnaire survey+interview by phone	Refused to answer or ignored the inquiry	Participation in the first project stage in early 2023
1	Institute of Mathematics of the National Academy of Sciences of Ukraine	x	x				
2	O. Ya. Usikov Institute for Radio Physics and Electronics	x	x				
3	Kherson State University	x	x				
4	Petro Mohyla Black Sea National University	x	x				
5	Zaporizhzhia National University	x	x				
6	S. Subbotin Institute of Geophysics	x	x				
7	Berdyansk State Pedagogical University	x	x				
8	State Biotechnological University	x					
9	Mariupol State University	x	x				
10	Kharkiv I.P. Kotlyarevsky National University of Arts	x	x				

##	Name of institution with damaged or destroyed research infrastructure	Filled-in questionnaire	In-depth interview	News report about the institution	Combined questionnaire survey+interview by phone	Refused to answer or ignored the inquiry	Participation in the first project stage in early 2023
11	Sumy National Agrarian University	x	x				x
12	V.N. Karazin Kharkiv National University	x	x				x
13	M. M. Hryshko National Botanical Garden of the National Academy of Sciences of Ukraine	x	x				x
14	State Archive of Kherson Region	x	x				
15	Dmytro Motornyi Tavria State Agrotechnological University	x	x				
16	Vasylivka Historical and Architectural Museum-Reserve "Popov Manor House"	x	x				
17	Institute of Radio Astronomy of the National Academy of Sciences of Ukraine	x	x				x
18	Institute for Problems of Cryobiology and Cryomedicine	x					
19	Institute of Climate-Smart Agriculture of the National Academy of Agrarian Sciences of Ukraine	x		x	x		
20	V. Bakul Institute for Superhard Materials	x					
21	State Specialized Enterprise ECOCENTRE	x					x
22	Institute for Safety Problems of Nuclear Power Plants	x	x				
23	Institute for Agricultural Microbiology and Agro-Industrial Production of NAASU	x	x	x			x
24	Institute of Applied Physics	x		x			
25	Odesa Polytechnic National University				x		
26	Ukraine Association of Biobank, Cell Biorehabilitation Institute			x	x		x
27	State Scientific Institution “Institute for Single Crystals” of the National Academy of Sciences of Ukraine			x		x	x

##	Name of institution with damaged or destroyed research infrastructure	Filled-in questionnaire	In-depth interview	News report about the institution	Combined questionnaire survey+interview by phone	Refused to answer or ignored the inquiry	Participation in the first project stage in early 2023
28	Institute of Dermatology and Venereology of the National Academy of Medical Sciences of Ukraine					x	
29	Danilevsky Institute of Endocrine Pathology of the National Academy of Medical Sciences of Ukraine	x	x	x			
30	National Scientific Center “Institute of Experimental and Clinical Veterinary Medicine”					x	
31	Taras Shevchenko National University of Kyiv					x	
32	State Enterprise Izyum Instrument-Making Plant					x	
33	State Enterprise “Production Association Yuzhny Machine-Building plant named after A.M. Makarov” (YUZHMASH)					x	

Annex 2. Collection of data to calculate losses and needs to recover the Ukrainian research infrastructure damaged during the war

Collection of data to calculate losses and needs to recover the Ukrainian research infrastructure damaged during the war

Thank you for your willingness to share your knowledge that will help define the methodology to calculate losses and evaluate needs to recover damaged and destroyed research infrastructure* as part of the Science at Risk project.

Our team has an additional task to improve questionnaires developed by the interagency working group for the assessment of the immovable property in higher education and research institutions that was damaged or

destroyed due to the armed aggression of the Russian Federation.

Your contribution will enable systematisation of this information for further analysis and definition of policymaking priorities of Ukraine regarding research infrastructure, and also will help find donors for restoration.

Our survey consists of two parts:
1. a questionnaire that will take around 4**0–60 minutes** to complete;
2. a 60-minute-long in-depth interview via video- or phone call whenever it is convenient for you.

The interview will be recorded for further encryption and analysis.

Below you will find the questionnaire consisting of five sections:
1. General information on research infrastructure
2. Information on damages
3. Calculation methods to assess damages
4. Formation of a budget for recovery and fundraising
5. Your experience of completing the questionnaire on damaged and destroyed property, developed by the interagency working group

To answer the second section of the questionnaire (information on damages), you will need information from your organisation regarding assessment of damages and losses. Answers for the third section of the questionnaire require involvement of a person responsible for the calculation and recording of losses inflicted. Questions in the fifth section are about your experience of completing the questionnaire sent by the interagency working group in July–August 2023.

Thank you for your time and participation!
Sincerely,
Maria Moskovko and Igor Lyman

Project: *Science at Risk*

Science at Risk is an expert community and web platform to help promote the expertise of Ukrainian scientists, find international partners and donors, and tell the stories of Ukrainian scholars in the times of war.

* In this survey, we use the following definitions:

Research infrastructure is a set of facilities, resources, and related services used by the research community to conduct top level research, covering essential scientific equipment, facilities, or sets of instruments tools, knowledge-based resources (collections, archives, depositories, or databases of scientific information), communication technology infrastructure (data grids, computers, software, and networks), and other structures of a unique nature. Research infrastructures may be local, virtual, or distributed (an organised network of resources), public or private. Research infrastructures can be a part of international networks (definition by: *Ministry of Education and Science*).

Intangible losses are loss of life and time, reputational damage, lost opportunities and potential, etc.

Indirect damages mean funds not received for scheduled contractual works that one failed to perform or could not perform in time due to a lack of electricity, water, or damaged equipment, or inability to maintain other statutory activities of institutions and organisations, such as providing services, promoting science, educational programmes etc.

I. General Information

1. Name of institution
2. Affiliation
3. Form of ownership
4. Type of research infrastructure (list multiple options, if any)
5. Does your research infrastructure have a special status?
6. Brief description of equipment or facilities
7. Research areas where this infrastructure is essential

II. Information on damages of research infrastructure

1. What type of damage was caused to your organisation's research infrastructure?
2. Is it possible to restore your research infrastructure?
3. What is the status of research conducted with the help of this infrastructure, as you fill in this questionnaire?
4. Cost of damage, in UAH
5. Amount needed for restoration, in UAH
6. Please specify whether the information on damages of your research infrastructure is posted on your organisation's webpage.
7. If yes, please provide a link to this information on your organisation's webpage.
8. Is this information available in English? If yes, please provide a link to this information.

III. Calculation methods to assess damages

1. Please describe the process used by your organisation to assess the damages.
2. Why and how was this calculation method chosen and by whom?
3. Does this method include intangible losses?
Intangible losses mean loss of life and time, reputational damages, lost opportunities or potential, etc.
4. If yes, what are they and how were they assessed?
5. Does this method include indirect damages?
Indirect damages mean funds not received for scheduled contractual works that one failed to perform or could not perform in time due to a lack of electricity, water, or damaged equipment, or inability to maintain other statutory activities of institutions and organisations, such as providing services, promoting science, educational programmes etc.
6. If yes, what are they and how were they assessed?
7. Please upload documents (if any) you used as guidance when assessing losses
If no uploading options are available, you can leave an access link in the next question.
8. Here you can leave a link to documents you used as guidance when assessing losses.

IV. Formation of a budget for recovery and fundraising

Fundraising is a process of attracting resources for project implementation.
1. Please describe the process used by your organisation to assess the cost of recovery.
2. Please provide the specifics of how exactly the cost of recovery was assessed:
3. Why and how was this method chosen and by whom?
4. Does your organisation receive aid for preservation/recovery of research of infrastructure?
5. If yes, and if this information is public, what kind of aid do you receive and who is the donor?
If this information is not available for the general public, please provide generalised answers, such as private foundations, individual donations, grants, charity donations, etc.
6. Does your organisation have a fundraising strategy?
A fundraising strategy is a formal or informal algorithm of actions to raise, use, and provide a report on usage of funds enabling the achievement of a specific goal for the development of research infrastructure.
7. If yes, please briefly describe this strategy.
8. What types of aid does your organisation require, besides direct financial aid?
Please select option(s) from the list or describe what else your organisation needs that is not on this list:
9. Is your organisation able to provide information on previous projects, audit reports

for the recent years, tax reports, or reports on usage of state funding?
10. Does your organisation have experience of this type of reporting?
11. If yes, please provide brief information on reporting mechanisms and experience.
12. Who is responsible for acceptance of donor funds in your organisation?
13. Does your organisation have a developed conflict of interest policy?
14. Please specify where the information on fundraising can be found on your organisation's webpage.
15. Does your organisation have an authorised point of contact who can communicate in English with the potential donors?
If yes, please provide contact details (name and email).

V. Your experience of completing the questionnaire on damaged and destroyed property, developed by the interagency working group

If you answered "no, haven't received and haven't filled in" in question 1 of this section, please skip to question 11.
1. Did your organisation receive an inquiry in July-August 2023 to fill in the table "Information on equipment/facilities of higher education/ research institutions, damaged or destroyed due to the armed aggression of the Russian Federation" and the file titled "Information on HEI/research institution immovable property damaged or destroyed due to the armed aggression of the Russian Federation"?
2. If yes, please rate from 1 to 5 the clarity of instructions and requirements for filling in the tables.
3. If your previous answer is 3 or lower, please specify what difficulties you have faced when filling in the tables.
4. What would you recommend adding/ removing/changing in the format and structure of the tables?
5. What would you recommend adding/ removing/changing in the content and relevance of questions?
6. Do you have any recommendations regarding frequency and intervals between inquiries on destruction or damages?
7. Do you have any recommendations regarding the process of receipt of such inquiries to fill in the tables?
8. Do you have any recommendations regarding the timeframes for filling in the tables?
9. Do you have any recommendations regarding the process of sending the answers?
10. Do you have any recommendations regarding feedback from the interagency group after you send back the filled in tables?
11. If you wish to add any information not contained anywhere else in this questionnaire but, in your opinion, important for our work, please specify here:
12. Please specify dates and time when you can meet for the second stage of this survey, the interview.

Annex 3. Questions for in-depth interviews

I. Calculation of losses and recovery needs

1. Please specify calculated losses in UAH
2. Please specify calculated costs needed for recovery, in UAH
3. Please describe the calculation process and method used by your organisation.
4. What are the challenges or issues you faced during calculation? Have you had any difficulties with the expert opinions? If yes, please describe. What could have

made this easier?
5. Have you taken into account intangible losses or indirect damages?

Intangible losses are loss of life and time, reputational damage, lost opportunities and potential, etc.

Indirect damages mean funds not received for scheduled contractual works that one failed to perform or could not perform in time due to a lack of electricity, water, or damaged equipment, or inability to maintain other statutory activities of institutions and organisations, such as providing services, promoting science, educational programmes etc.

1. Have you used any unique methodology to calculate any type of loss that is not commonly known, and would you recommend it to other institutions of Ukraine?

1. Based on your organisation's experience in assessing damages and destroyed property, do you have any recommendations for the government, other organisations with damaged or destroyed research infrastructure, or the international community etc. regarding:

a) the process of assessing damages;
b) strategy for recovery and development of damaged or destroyed research infrastructure?

II. Fundraising
1. How does your organisation raise funds to recover damaged or destroyed RI?
2. What types of support does your organisation require, besides direct financial aid?

Please choose option(s) from the list or specify what else your organisation needs that is not on this list:
- Capacity development for scientists
- Grant application writing skills
- English language courses
- Networking with international partners
- Information on technology transfer and commercialization of research

1. If SAR finds potential donors, is your organisation able to provide information on previous projects, audit reports for the latest years, tax reports, or reports on usage of state funding?

III. Experience of completing the questionnaires on damaged and destroyed immovable property, developed by the interagency working group
1. If you received and completed the questionnaires:

- What would you recommend adding/removing/changing in the format and structure of the tables?
- What would you recommend adding/removing/changing in the content and relevance of questions?
- Do you have any recommendations regarding frequency and intervals between inquiries on damages and destruction of property?
- Do you have any recommendations regarding the process of receiving inquiries to fill in the tables?
- Do you have any recommendations regarding the timeframes to fill in the tables?
- Do you have any recommendations regarding the process of sending the answers?
- Do you have any recommendations regarding feedback from the interagency group after you send back the filled in tables?

12. Other than completing a questionnaire, what means would be more effective and less time-consuming for your organisation to inform the governmental authorities on your damaged or destroyed property?
- A reporting system already available in practice or work regulations
- An online system that can be periodically

updated
- A focus group
- Personal contact (phone, video call)
- Filling in a (completely or partially) public table on an online resource

13. Is there anything we should have asked during this interview but haven't?

Other suggestions, additions, recommendations, advice etc.?

Annex 4. Damages/destruction (general)

Information on immovable property of higher education/research institution, damaged or destroyed due to the armed aggression of the Russian Federation
as of _____ 2023 (date)

##	Name of organization – balance sheet holder of property	Name of immovable property object	Type of immovable property (building/ engineering facilities)	Purpose of building (research/ other activities)	Location	Inventory number	Total area, square meters.

##	Residual value of property, UAH (based on report of evaluation/ inspection report/ inventory report (if any))	Market value of property, UAH	Registration number of property in the State Register of Real Property Rights	Results of criminal act (damaged/ destroyed)	Availability for target usage (possible/ partially possible/ impossible	Type of works needed for restoring operation of property (routine repair/ major overhaul / reconstruction/ building)	Estimated costs for recovery, UAH

Annex 5. Damaged/destroyed equipment (general)

Information on equipment/facilities of higher education/research institution, damaged or destroyed due to the armed aggression of the Russian Federation
as of _____ 2023*

##	Name of organization – balance sheet holder of property	Name of immovable property object	Type of immovable property (building/ engineering facilities)	Purpose of building (research/ other activities)	Location	Inventory number	Total area, square meters.

##	Residual value of property, UAH (based on report of evaluation/ inspection report/ inventory report (if any))	Market value of property, UAH	Registration number of property in the State Register of Real Property Rights	Results of criminal act (damaged/ destroyed)	Availability for target usage (possible/ partially possible/ impossible	Type of works needed for restoring operation of property (routine repair/ major overhaul / reconstruction/ building)	Estimated costs for recovery, UAH

* for the equipment/facilities costing ≥ UAH 20,000

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